

ORBITAL-HAR

Offline on-board procedure supervision for BAS experiments

Fully offline

Edge AI

Explainable & auditable

PROBLEM STATEMENT · SIH26174

AI Human Activity Recognition for On-board BAS Experiments

ORGANISATION

ISRO · Dept. of Space

THEME · CATEGORY

Space Technology · Software

TEAM

Hashira

TEAM ID

(as registered on portal)



Proposed Solution

THE IDEA IN ONE LINE

A fixed payload camera watches an astronaut run an experiment — the system recognises each action, prompts the next step, voice-alerts on a skip or wrong order, and emits a few KB of verified telemetry instead of a video downlink.

HOW IT ADDRESSES THE PROBLEM



Lightweight downlink

A few KB of hash-chained step telemetry replaces high-bandwidth video — provable status without real-time comms.



Reasoning-first, never guesses

A deterministic state machine decides; below confidence it reports UNVERIFIED rather than a confident wrong verdict.



Orientation-agnostic

The input frame is canonicalised to the payload rack (ArUco), so a floating or inverted astronaut is still recognised.

INNOVATION & UNIQUENESS



Procedure-as-config

Every experiment is a YAML file, not code — a new experiment is a new file, engine untouched.



Event-bus seam

Perception and reasoning are decoupled: record once, replay 1000× — and live demo insurance if a camera fails.



Explainable & auditable

Deterministic verdicts plus an append-only, hash-chained log make every decision defensible.

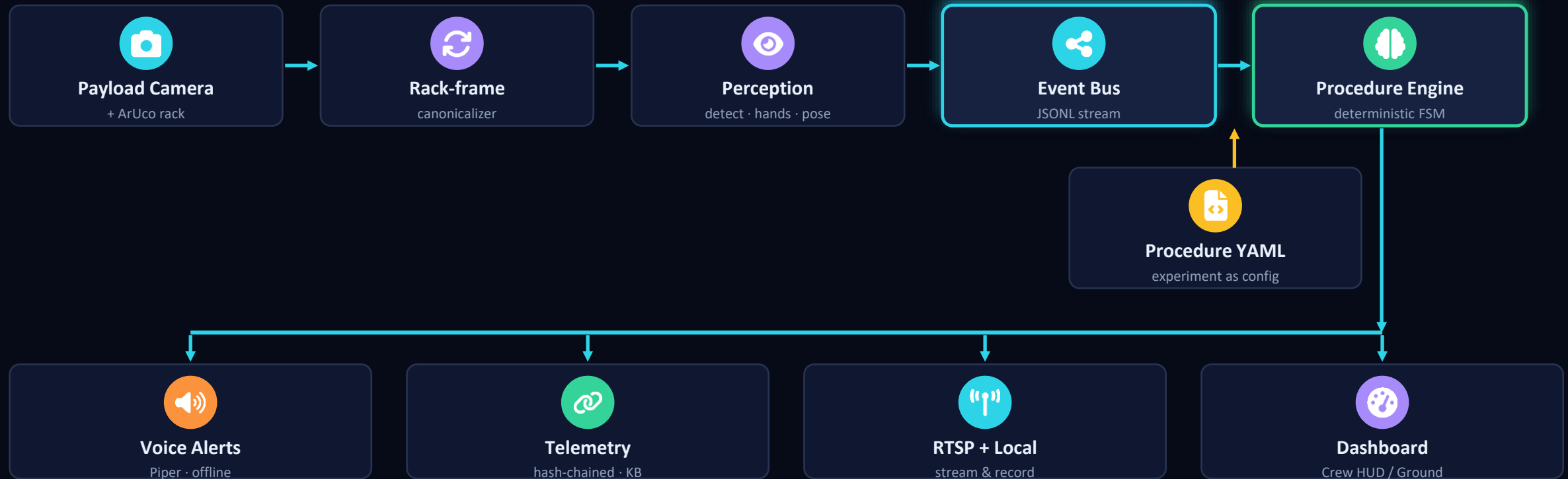
Technical Approach

PERCEPTION YOLO11 · MediaPipe Hands · YOLO11-pose · OpenCV ArUco

REASONING Pydantic + YAML procedures · deterministic state machine · 7 predicate evaluators

RUNTIME Piper offline TTS · FastAPI + WebSocket · React + Vite (bundled) · FFmpeg + MediaMTX

EDGE Jetson Orin Nano · TensorRT export · Blender synthetic data · zero network at runtime



Deterministic engine emits — fire-and-forget: losing any output never stops supervision.

Feasibility & Viability

ALREADY DE-RISKED

73

tests passing

Reasoning spine proven end-to-end with no camera and no models.

LIVE

webcam demo today

A pretrained YOLO stand-in already feeds the real engine — HUD + voice.

~2,800

frames to train

≈2,000 synthetic + 800 real; fixed camera, <12 classes = small-data regime.

RISKS → MITIGATIONS

Dataset generation is the critical path



Blender auto-labelled synthetic data, randomised across the full orientation sphere

Pose models fail on inverted bodies



Canonicalise the input frame, not the output pose — model sees an upright person

Real-time latency in the alert path



Pre-synthesize voice prompts to WAV; TensorRT export on the Jetson

Camera fails during the live demo



Event-bus replay of a recorded session — the demo keeps running

Impact & Benefits



GB → KB

DOWNLINK PER SESSION

Verified, hash-chained step telemetry replaces raw video — an orders-of-magnitude saving on a bandwidth-restricted space link.



Mission autonomy

Protocol-perfect science execution beyond real-time ground support — the core need of the PS.



Safety & trust

Deterministic, tamper-evident audit of every step reduces crew error and supports post-mission review.



Reusable platform

Procedure-as-config generalises to surgical, industrial, hazardous-site and training tasks.



National capability

Indigenous, fully-offline edge-AI aligned with India's BAS and lunar programme.

Research & References

-  SIH 2026 · SIH26174 — AI HAR for On-board BAS Experiments, ISRO / Dept. of Space sih.gov.in/sih2026PS
-  Ultralytics YOLO11 / YOLO-World — detection & open-vocabulary pre-labelling docs.ultralytics.com
-  Google MediaPipe Hands — real-time hand landmarks & hand-object interaction mediapipe.dev
-  OpenCV ArUco — 6-DoF fiducial marker pose for rack-frame canonicalisation docs.opencv.org
-  Piper TTS (Rhasspy) — fully offline neural text-to-speech for voice alerts github.com/rhasspy/piper
-  Orientation-agnostic 3D Human Mesh Recovery (HMR) — rack-relative body tracking [research literature](#)
-  Blender bpy — scripted synthetic-data generation with automatic labels docs.blender.org

The reasoning spine is done and tested — the trained AI model is the next milestone. **ORBITAL-HAR turns any procedure into a supervised, auditable, offline mission asset.**

Team Hashira

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